

EE 601L: Neural Networks and Deep Learning

Lab-1

Pakalapati Naga Srinivasa Varma

January 12, 2026

1 Dataset

The Iris dataset contains 150 samples from three flower species so this is a multi-class classification with 3 classes.

2 Analysis and Observations

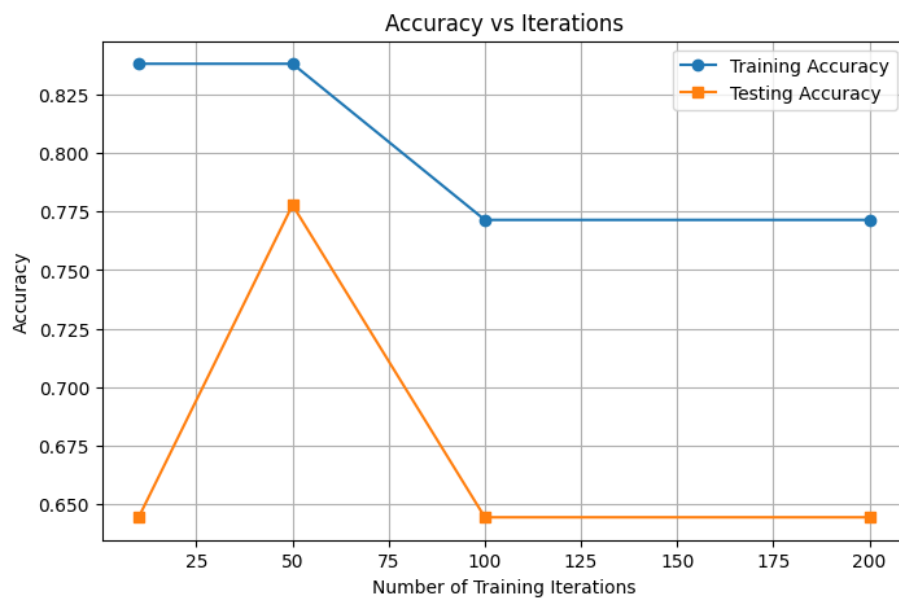


Figure 1: Plot of Training and Test Accuracy vs Iterations

When predicted on the training data, the accuracy was high, about 0.8380 and remained constant until 50 iterations. However, after 50, it decreased and reached 0.7714 and remained constant from 100-200 iterations.

This might be because until 50, the model had made nice decision boundaries but since the data is non-linearly separable, the misclassifications still existed. So, after 50 iterations, every new correction(update of weights) led to more misclassifications which further decreased the accuracy.

Similarly on testing data, the accuracy reached a peak value of 0.777 on 50 iterations and declined after that due to the same reason of more misclassifications occurring after changing the weights.

These observations can be seen in the decision boundaries plotted below:

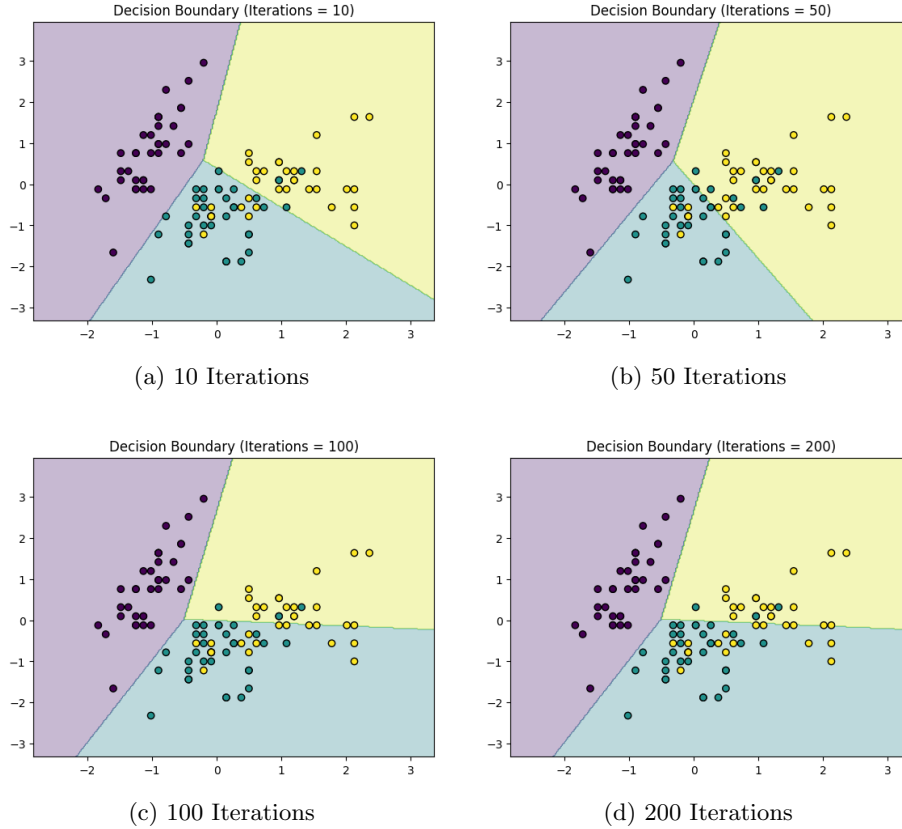


Figure 2: Decision Boundaries

3 Limitations

- 1. Works only for Linearly Separable data:** The Perceptron model cannot handle linearly separable datasets. Iris dataset is non-linearly separable.
- 2. Convergence Issues:** The Perceptron converges only when the classes can be linearly separable. So, for non-linearly separable data, misclassifications do not decrease so it just keeps oscillating.
- 3. Limited to Decision Boundaries:** Decision boundary is always a straight line, plane, or hyperplane, depending on dimensions.

4 Improvements which can be made

- 1. Use More Features from the Dataset:** Instead of using only two features, all four features of the Iris dataset can be used which will improve class separability and classification accuracy.
- 2. Use a Multi Layer Perceptron:** We can stack the single perceptron and add more layers which will help us overcome the limitations of linear separability and complex decision boundaries can be made.
- 3. We can use Kernel Methods:** This will transform the non-linearly separable data in the current dimension to a linearly separable data in a higher dimension.

5 Conclusion

In this experiment, a multi-class Perceptron was implemented and evaluated on the Iris dataset using only two features and the effect of the number of training iterations on both training and testing accuracy was analyzed.

It was observed that the accuracy increased until 50 iterations and decreased afterward, which is due to the non-linearly separable nature of the data. Since the Perceptron does not minimize a loss function and updates weights only on misclassifications, further iterations led to unstable decision boundaries and reduced accuracy.

The decision boundary plots clearly show how the boundaries change with iterations and explain the variation in accuracy.

Overall, this experiment helped in understanding the working, limitations, and behavior of the multi-class Perceptron model.